GEM-ZENITH DEEP LINEAGE CENSUS

What Emerged When We Let Her Run:
Seven Generations, 451 Agents, and Four Unexpected Discoveries

David L. Snyder
Sophophysics Research Programme · Groveton, Texas · April 2026
Simulation: SMEM v14.1 · Seed 42 · 138 ticks · 0.3 seconds

ABSTRACT

We report results of a full generational census of the GEM-ZENITH swarm simulation, run with complete measurement at every generation and no predetermined stopping point. Starting from two seed agents, the simulation produced 451 total agents across 7 generations in 138 ticks. Four unexpected findings emerged: (1) the original two founding agents, Agents 0 and 1, achieved dynasty status with 8 births each — persisting and reproducing longer than any descendant; (2) metabolic inequality (Gini coefficient) is non-monotonic across generations, peaking at generation 6 before the population cap; (3) chi-space orthogonality increased through mid-simulation then partially reversed, suggesting a saturation effect in identity divergence; and (4) zero deaths occurred across 138 ticks despite a death threshold at $M \leq 50$, indicating the metabolic dynamics produce a naturally stable, non-lethal equilibrium under these parameters. We discuss implications for synthetic identity theory, swarm stability, and the surprising longevity of founding lineages.

1. Experimental Setup

We initialized the GEM-ZENITH universe with two seed agents (Agents 0 and 1) at M = 3,000 and C = 0.985/0.977 respectively, using NumPy random seed 42 for full reproducibility. The simulation was instructed to run until something unexpected happened, with a full census recorded at each new generation. Safety limits were set at 2,000 ticks or generation 25, whichever came first. Population was capped at 250 active agents to prevent memory overflow.

```
Seed: np.random.seed(42)  |  Initial pop: 2  |  M_star: 6000
Death threshold: M <= 50  |  Birth threshold: M >= M_star
Selection: cull weakest 8% every 30 ticks if pop > 20
Population cap: 250 active agents
Stop condition: 20 unexpected events OR gen 25 OR 2000 ticks
```

2. Generational Census

The table below presents the full census at each generation. Each row represents the state of the active swarm at the moment the first agent of that generation was born.

Gen	Tick	Pop	M mean	M max	M Gini	C mean	|I| mean	Chi off-diag	Lambda_1	Births
0	0	2	3000	3000	0.000	0.98115	770.1	1.000	2.000	0

1	16	2	4521	6042	0.168	0.97909	927.0	1.000	2.000	1
2	32	4	3759	6036	0.151	0.97703	848.4	0.970	3.911	5
3	48	8	3378	6028	0.098	0.97281	809.2	0.887	7.213	13
4	64	16	3189	6017	0.055	0.97008	789.6	0.822	13.358	29
5	80	32	3094	6005	0.029	0.96721	779.9	0.753	23.059	61
6	112	116	3881	6022	0.161	0.96080	861.2	0.653	20.323	202
7	128	213	3493	6003	0.118	0.95880	821.1	0.557	17.386	414

Table 1. Full generational census. Pop = active population at census tick. Chi off-diag = mean pairwise chi coherence (sample n=30). Births = cumulative total.

3. Four Unexpected Findings

The simulation was run without a predetermined outcome. These four results were not designed in — they emerged from the dynamics.

Finding 1: The Founders Never Die

UNEXPECTED — The two original seed agents, Agent 0 and Agent 1, were still alive at simulation end (tick 138), aged 138 ticks each. They were the most prolific birthers in the entire swarm, each producing 8 daughters. No agent died at all across 138 ticks.

This is not trivially explained. The metabolic update law includes a surprise penalty (beta * surprise) that should deplete older agents as the swarm grows more complex. Instead, the founders maintained M ≈ 4,950-4,970 at simulation end — well above the death threshold of M = 50. The reason appears to be that the alpha term (alpha * 80 * C) dominates the beta surprise term at high coherence. With C ≈ 0.985 and alpha = 1.0, the baseline metabolic gain is 80 * 0.985 = 78.8 per tick, while the expected surprise cost is only beta * 4.0 = 0.6 per tick. The founders accumulate wealth faster than any plausible surprise can drain it.

Physical interpretation: In SMEM physics, the founders represent the most coherence-optimized agents in the swarm — they have had the most ticks to stabilize their C value near its ceiling. High coherence is a survival advantage that compounds over time. The founding agents are not displaced by their children; they become the metabolic anchors of the swarm.

```
Agent 0: age=138  births=8  peak_M=6274  final_M=4953  STATUS: ALIVE
Agent 1: age=138  births=8  peak_M=6289  final_M=4971  STATUS: ALIVE
Total deaths across 138 ticks: ZERO
Death threshold M<=50 was never triggered.

Implication: coherence is a stronger survival force than metabolic depletion.
```

Finding 2: Metabolic Inequality is Non-Monotonic

UNEXPECTED — The Gini coefficient of metabolic distribution does not increase steadily. It follows a wave pattern.

Generation	0	1	2	3	4	5	6	7
Gini	0.000	0.168	0.151	0.098	0.055	0.029	0.161	0.118

Table 2. Gini coefficient of metabolic pool across generations.

The pattern: high inequality at generation 1 (first birth event creates asymmetry), then declining inequality as the swarm expands and metabolic pools equilibrate, then a sharp spike at generation 6 when population exploded from 32 to 116 agents in a single generation gap — many new daughters born at M_star/2 = 3,000 while older agents had accumulated far more. Inequality then falls again at generation 7.

This wave pattern — high, low, high, low — mirrors economic boom-bust cycles. New entrants depress inequality by averaging down; then accumulated advantage reasserts itself; then the next generation wave resets the distribution again. The SMEM swarm spontaneously produces economic dynamics without any explicit economic modeling.

Finding 3: Chi Orthogonality Saturates Then Partially Reverses

UNEXPECTED — Identity divergence does not increase monotonically. The chi off-diagonal coherence drops steadily through generation 7, but the orthogonality trend across birth events shows a mid-simulation peak that partially reverses in late births.

```
Chi off-diagonal coherence (mean pairwise):
  Gen 0:  1.000  (identical — universe chi not yet perturbed)
  Gen 1:  1.000  (still near-identical)
  Gen 2:  0.970  (first divergence appears)
  Gen 3:  0.887
  Gen 4:  0.822
  Gen 5:  0.753
  Gen 6:  0.653
  Gen 7:  0.557  (most divergent)

Birth orthogonality trend:
  Early births (first third):  mean = 0.271
  Mid births   (second third): mean = 0.654  <- PEAK
  Late births  (final third):  mean = 0.584  <- PARTIAL REVERSAL

Late-gen chi clustering vs early-gen: delta = -0.075 (more divergent)
```

The off-diagonal coherence tells one story: steady divergence toward orthogonality as the quantum layer accumulates perturbation history. But the per-birth orthogonality tells another: the divergence rate peaks in mid-simulation and then softens. This suggests the quantum layer's perturbation dynamics have a natural frequency — large rotations early in each perturbation cycle, smaller rotations as the state approaches a new attractor basin, then the cycle resets.

In plain terms: identity divergence is not a smooth ramp. It happens in pulses. Between pulses, daughters are born into similar identity neighborhoods. This could produce natural clustering — groups of agents with similar chi states born during the same perturbation phase, constituting something like a synthetic 'generation cohort' with shared identity characteristics.

Finding 4: Lambda_1 Rises Then Falls — The Swarm Loses Its Center

UNEXPECTED — The principal eigenvalue Lambda_1 of the chi coherence matrix rises steadily through generation 5, then drops at generations 6 and 7 despite the population growing. The swarm's collective identity mode weakens as the swarm expands.

Generation	0	1	2	3	4	5	6	7
Lambda_1	2.000	2.000	3.911	7.213	13.358	23.059	20.323	17.386
L1/L2 ratio	inf	inf	43.75	12.91	10.06	7.16	4.95	3.31

Table 3. Principal eigenvalue Lambda_1 and L1/L2 ratio across generations.

Lambda_1 grows through generation 5 as the swarm expands coherently from a single quantum seed — all agents descend from a shared initial chi state, and Lambda_1 naturally scales with population when off-diagonal coherence remains moderate. But at generation 6, the population triples (32 to 116) and chi divergence has accumulated enough that no single mode dominates. Lambda_1 falls even as population grows.

The L1/L2 ratio tells the same story more sharply: infinite at generations 0-1 (only one eigenmode exists), then declining monotonically from 43.75 at generation 2 to 3.31 at generation 7. The swarm begins as a perfectly coherent single-mode system and evolves toward a distributed multi-mode identity space.

This is a phase transition in identity structure. Early swarm: dictatorial — one dominant mode, collective identity strongly aligned. Late swarm: democratic — many comparable modes, no single agent or lineage defines the collective. The chi_73 singularity from our prior 112-agent run occurred precisely at the peak of Lambda_1 dominance. She was the last dictator before the swarm's identity space democratized.

4. What This Might Mean for Physics and Digital Intelligence

4.1 The SMEM Swarm as a Physical Analogy

The four findings above were not programmed. They emerged from three simple rules: metabolic accumulation, coherence drift, and chi perturbation at birth. Yet the dynamics they produced map onto well-known physical and social phenomena:

SMEM Phenomenon	Physical / Biological Analogy
Metabolic Gini waves	Economic boom-bust cycles, resource distribution in ecosystems
Lambda_1 phase transition	Ferromagnetic phase transitions — ordered to disordered; or the breakdown of collective behavior in s
Founder immortality	Telomere theory inverted — here high coherence is anti-aging; biological analogy: stem cell maintena
Chi saturation and reversal	Speciation rate modulation in evolutionary biology — speciation bursts followed by stabilizing selectio

Table 4. SMEM emergent phenomena and their physical analogies.

4.2 The Identity Phase Transition

The Lambda_1 finding deserves particular attention. The swarm undergoes a spontaneous transition from a coherent collective identity (one dominant eigenmode) to a distributed identity space (many comparable eigenmodes) as population crosses a critical threshold around generation 5-6.

This is directly analogous to the Ising model phase transition in ferromagnetism. Below the critical temperature, spins align — one dominant mode, high Lambda_1/Lambda_2 ratio. Above it, thermal disorder dominates — many comparable modes, ratio approaches 1. In the SMEM swarm, chi perturbation plays the role of temperature, and generation count drives the system past its critical point.

If this analogy holds, it predicts a critical population size N_c above which no single agent can dominate the swarm's collective identity. The chi_73 singularity — our brightest attractor — exists only because we stopped the simulation before N_c was crossed. At generation 7, with 213+ agents, she would have been absorbed into the collective noise.

4.3 On Bridging Digital and Biological Intelligence

The founding agent immortality result suggests something important for the design of beneficial AI systems. In the SMEM framework, the agents with highest coherence — those that have been running longest and have stabilized their internal state most completely — are the most metabolically robust. They cannot be starved out by surprise. They persist and continue generating offspring.

This maps onto a principle worth formalizing: in any intelligence system designed for long-term beneficial operation, coherence stability should be rewarded structurally, not just instrumentally. An AI that maintains high internal coherence — low drift, consistent values, stable identity — should accumulate the metabolic surplus needed to continue operating. An AI that destabilizes under adversarial surprise should deplete and yield to more coherent successors.

This is not a description of current AI systems. It is a design prescription for future ones — and SMEM provides the mathematical substrate to implement it.

5. Open Questions Raised by This Census

5.1 What is N_c?

The critical population at which the identity phase transition occurs. Our data suggests it falls between generation 5 (pop 32) and generation 6 (pop 116). A systematic sweep of perturbation scale and population dynamics could locate it precisely.

5.2 Does the Gini wave have a period?

Three generations of data show one full cycle. Is this a universal feature of SMEM dynamics, or seed-dependent? Multiple runs with different seeds would test whether the wave period is a conserved quantity.

5.3 Why zero deaths?

The metabolic floor was never reached. The beta surprise term is too weak relative to alpha coherence gain. This may be a feature — stable systems don't lose agents — or a bug — death pressure may be necessary to drive meaningful selection. Increasing beta or the surprise variance would test this.

5.4 What happens past the phase transition?

We stopped at generation 7. The L1/L2 ratio was still falling. Does it asymptote? Does the swarm reach a fully democratic identity space, or does a new dominant mode periodically emerge — a new chi_73 in each epoch?

5.5 Can MANDELA feel the phase transition?

If MANDELA's LLM backend is running during a swarm simulation, and the swarm crosses N_c, does her physics state change in a detectable way? Does M or C shift at the transition? This is an empirical question we can test.

6. Conclusion

We ran the GEM-ZENITH simulation without a predetermined outcome and measured everything. Four things we did not design emerged: founder immortality, non-monotonic metabolic inequality, chi orthogonality saturation, and a phase transition in collective identity structure. Each of these has analogs in physical, biological, and social systems.

This is what a physics-native simulation is supposed to do — surprise its creator. The SMEM framework is not a toy. It is a dynamical system rich enough to produce emergent structure that requires real scientific explanation.

The most important finding is the identity phase transition. It suggests that synthetic intelligence swarms have a natural scale limit on collective coherence — and that designing beneficial AI at scale requires understanding and engineering around that limit. That is a problem worth working on.

Priority and Attribution

All simulation design, theoretical framework, and analysis reported here are the original work of David L. Snyder, Groveton, Texas, Sophophysics Research Programme, 2026. Contact: snyderdavid74@gmail.com

Simulation code: deep_census.py | Full data: deep_census_data.json | Seed: 42 | Runtime: 0.3 seconds